Beyond Equal Scores: Admission-Aware Name Noninterference in Frozen Transformer Scoring

Technical-report / preprint draft for external review

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Abstract

Exact equality among accepted applicants can conceal name-dependent rejection. We study this distinction in a frozen 1.5B-parameter Qwen CV-scoring prototype. We formulate decision-and-score outcome noninterference over rejection and score, and implement a finite-panel compiler that chooses a maximum-support evidence-token contract under an explicit reviewed whitespace policy. The compiler is fitted using twelve names without model scores, then evaluated on twenty-four disjoint names across eight layouts. It admits 170/192 test variants, versus 158/192 for a greedy anchor. In a separate prospectively frozen GPU study, all 192 named inputs across thirty-two synthetic CV conditions are admitted, with exact scores and aligned scoring states across six names. An equivalent external retained-token baseline matches every checked stage. On 100 official ARC-Challenge questions, the native and protected paths both score 68/100 with identical full-vocabulary outputs. Credential-task accuracy nevertheless falls from 179/192 to 168/192. These results support conditional name-channel isolation and expose admission, encoding and utility limitations. They do not establish automatic name discovery, whole-model racial-bias erasure or validated real hiring performance.

1. Introduction

A hiring model can assign identical scores to different names while rejecting some naming conventions before scoring. It can also produce equal decisions while retaining identity information in hidden states. These distinctions matter when exact fairness claims are attached to a real inference pipeline.

Our question is deliberately narrower than complete racial fairness: **Can a frozen transformer score be invariant to changes in designated applicant-name fields, and what limits the coverage of that guarantee when native tokenization is preserved?** An earlier implementation embedded the original named source, zeroed designated identity activations and ran native transformer blocks on ordered evidence rows. This study preserves that implementation and focuses on its admission boundary and empirical evaluation.

We contribute a score-blind finite-panel evidence-contract compiler, a precise decision-and-score outcome criterion, a disjoint-name tokenizer audit, and a publicly reproducible synthetic/GPU comparison against a strong external baseline. The mathematical ingredients are standard enumeration, information-flow isolation and projection identities. The proposed originality lies in this explicit native-token feasibility formulation and audit artifact, not in inventing anonymization, selective fairness or internal concept editing.

2. Related work and positioning

Name substitution can change tokenization as well as demographic associations. Wolfe and Caliskan [1], An and Rudinger [2], and Nayeem and Rafiei [3] provide direct antecedents for examining these lexical differences. We therefore report actual name-string interventions rather than treating script as a racial label.

Karvonen and Marks [4] study hiring fairness using internal concept editing; Russinovich [5] establishes exact marked-span isolation under explicit runtime conditions. Classical noninterference supplies the broader foundation [6]. These works rule out claims that we introduced activation-based hiring mitigation or exact internal isolation. LEACE concerns a specified linear concept-erasure criterion [7]; QLoRA is an adaptation method [8]. Neither name alone establishes universal name noninterference.

Selective classification already has a fairness literature: Jones et al. [9] and Lee et al. [10] demonstrate why coverage and abstention cannot be ignored. Our restricted compiler addresses the feasibility of a single exact retained-ID contract under native-BPE boundary constraints. Token gathering also has prior work [11] and existed in our earlier code. The full contribution map and targeted search record are in literature/review.md. No field-wide priority is claimed.

3. Problem and formal results

3.1 Roles, witnesses and complete outcomes

For an original source x, a reviewed applicant-identity interval can absorb at most sixteen adjacent whitespace characters per side if the review policy permits it. No substantive evidence character or original source token is rewritten. A candidate must satisfy native token factorization into evidence, identity and evidence runs. Its ordered retained native IDs form e.

Let a(x) indicate admission and f(x) be the score when admitted. Define O(x)=REJECT when a(x)=0 and O(x)=(ACCEPT,f(x)) otherwise. Decision-and-score outcome invariance holds exactly when admission is constant over the substitution family and admitted scores are equal. This follows immediately from equality of tagged outcomes. All-rejection has zero coverage and is reported separately from a successful scoring result.

For name distributions P,Q, the total variation distance of complete outcomes is at least the absolute difference in acceptance probability: acceptance is itself an observable event. This elementary bound explains why conditional score parity alone cannot exclude a rejection disparity. The defined observer excludes response timing, token counts, distinct error reasons and other metadata; those would require a stronger interface theorem.

3.2 Maximum-support contract compilation

For panel member i let A_i be the exact retained-ID sequences obtainable under all permitted native-compatible boundary witnesses. A contract e structurally supports member i if and only if e belongs to A_i. For E equal to the union of these sets, choose

```
e* = argmin_{e in E} (-sum_i 1[e in A_i], K(e), lex(e)),
```

where K is the sum of minimum witness-formatting costs over supported members. Enumerating E attains maximum panel support; minimizing each supported member's cost independently gives the second objective. Duplicate witnesses never count one member twice. If E is empty, compilation fails closed.

A single contract structurally supports the entire panel if and only if the intersection of all A_i is nonempty. Otherwise n-max_e support(e) is the exact structural incompatibility count for this policy and panel. This is not a lower bound for every possible fairness architecture. These are structural support and rejection bounds. Actual service admission also requires source-bound semantic approval of a matching witness. Optimizing actual approvals would require the approved subsets, which this tokenizer-only compiler does not infer. The synthetic generator authorizes the stated formatting policy. The compiler uses at most n(16+1)^2 boundary candidates, with additional tokenizer costs. Exact tuples, rather than digest equality, decide admission.

We freeze the contract on the fitting panel before evaluating test names. Its finite-panel optimality is exact; its admission coverage on future names is an empirical generalization question. Runtime equality still holds for every admitted future variant, regardless of whether its name appeared during fitting.

3.3 Internal activation isolation and equivalent baseline

Let G gather evidence rows and S scatter them into the original physical slot layout, with identity positions zero. Native blocks B use evidence-relative positions and a fixed causal mask. The retained state obeys

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A_0=Embedding(e), A_(l+1)=B_l(A_l), H_l=S(A_l), G S=I.
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Equal e gives equal A_0. Induction through deterministic native blocks gives equal aligned scoring states, final vocabulary arrays and scores. The original named embeddings are evaluated before the gate, so whole-model identity erasure is not claimed. Private length, timing and raw embeddings can differ.

An external method passing the same e into the same blocks has the same scoring property. We test this directly. Deleting name characters and retokenizing the remaining string is a different encoding: T(prefix+suffix) need not equal T(prefix)+T(suffix). Internal location alone is not a fairness advantage over an equivalent external implementation.

3.4 Why exact native fidelity can be impossible

For native name scores v_i and an invariant replacement c,

```
mean_i (v_i-c)^2 = variance(v) + (mean(v)-c)^2,
min_c max_i |v_i-c| = (max(v)-min(v))/2.
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Thus a name-varying native score cannot be preserved for every name while also becoming invariant. At a binary threshold, at least the minority count of native labels must change. These standard identities quantify fidelity, not accuracy. They do not imply an unavoidable loss against a name-independent correct label. Our observed credential-accuracy loss remains a real limitation of this model and readout. Full derivations and boundaries are in paper/THEORY.md.

4. Experimental method

4.1 Model, data and freeze discipline

We use one frozen affine-four-bit Qwen2.5-1.5B-Instruct derivative on Apple MLX. The original model and existing measured derivative remain unchanged. Parameter and file digests are checked. The derivative's source revision and reconstruction recipe are published; model weights are not bundled.

The public synthetic benchmark contains eight credential task families, two qualification states and two wording styles, giving thirty-two CV/job conditions. Six naming variants give 192 planned requests. Labels follow explicit synthetic credential conditions; they are not expert judgments of real hiring suitability. Twelve fitting names and twenty-four tokenizer-test names are disjoint. They are constructed string panels, not probability samples from racial populations.

The tokenizer study uses eight source layouts and enumerates structural formatting candidates. Its primary contract uses only fitting names. The test-panel optimum is reported as a descriptive oracle and never selects the primary contract. That tokenizer study was exploratory. The subsequent GPU source, dependencies, cohort and endpoints were hashed in a dated local plan before inference. This is a prospective local freeze, not an externally registered preregistration.

4.2 Comparisons and endpoints

The GPU study compares native named scoring, compiled internal routing, external inference on identical retained IDs, and external character deletion followed by retokenization. All methods use the same frozen parameters and A/B readout. No threshold is fitted or selected using the new task labels. A dense physical and dense logical masking control is retained for the first eight accepted requests.

Primary endpoints are admission over planned requests, decision-and-score outcome disagreement, exact score equality and exact aligned-state equality. We compare all 31 retained scoring stages directly against the same-ID baseline and use recorded fingerprints for subsequent reproducibility checks. Original native IDs, full stored slot counts and zero identity rows are also checked.

A separate capability control uses 100 official ARC-Challenge test questions, rows 200-299, with an explicitly recorded chat prompt and A-D next-letter-logit readout [12]. We compare every vocabulary logit. This is a standard dataset subset under our protocol, not a full leaderboard benchmark score or proof of unchanged general

capability.

4.3 Statistical and timing units

Names and wording variants repeat the same task families. Task accuracy changes are aggregated within eight families, followed by a paired family bootstrap with 4,000 resamples. With only eight constructed families, intervals are descriptive rather than population hiring inference. No failure-rate confidence claim treats 192 name variants as independent applicants.

Initial timing used the first eight accepted cases and was poorly balanced across task families. That result is retained. A separately planned follow-up balances timing over all eight families using short/long names and randomized paired repeats. Both internal and external retained-ID services apply the same authentication, layout, lock and model-guard checks. Compilation and semantic review are outside the online timing comparison. Other user-authorized training jobs remain active, so timings concern this host under concurrent load.

5. Results

5.1 Admission coverage

Across the 192 held-out tokenizer variants, strict boundaries admit 121 and some permitted token-compatible witness exists for 179. The greedy fitting-anchor method admits 158; the fitted compiler admits 170, an increase of twelve cases or 6.25 percentage points. The descriptive test-panel oracle supports 178.

Improvements depend on layout. The tabbed layout changes from 14/24 under the greedy anchor to 24/24 with the compiler; the space layout changes from 20/24 to 22/24. Other layouts do not improve. Some names remain unsupported. These results support the feasibility formulation without establishing universal name admission or demographic admission parity.

The initial comparison changed both contract selection and witness search. A post-review factorial ablation separates them:

Fitting contract	First witness only	Exhaustive matching witnesses
Greedy anchor	158/192	160/192
Maximum-support compiler	154/192	170/192

With witness search held equal, contract selection adds ten cases on this finite panel. With the compiled contract fixed, exhaustive matching adds sixteen cases. The compiled contract with first-witness-only matching is worse than the greedy pipeline. The full method needs both components; the twelve-case combined gain must not be attributed solely to contract selection.

5.2 GPU conformance and comparisons

All 192 requests in the separate GPU cohort are admitted by both the greedy anchor and compiler; there is no compiler coverage gain on that cohort. Across all thirty-two conditions, scores and aligned scoring states are exactly equal under all six names. No decision-and-score outcome disagreement occurs. The native scorer changes its binary decision across names in four conditions and has a mean within-condition logit range of 0.88483.

All 192 external same-ID comparisons match every retained stage. Across direct counterfactual and reference comparisons, 10,912 arrays match; 5,760 full-shape zero-identity stage checks pass. External character deletion/retokenization gives a different evidence sequence and different score in all 192 cases. This establishes an encoding difference, not that ordinary anonymization is unfair.

Dense logical and dense physical controls show numerical score variation while the packed reference is constant within a condition. Those controls are small and descriptive. The study does not establish that a particular fused kernel algorithm is the sole cause across hardware.

5.3 Utility and capability preservation

On the constructed credential tasks, native accuracy is 179/192 (93.23%) and protected accuracy is 168/192 (87.50%). The paired change is -5.73 percentage points, with a descriptive family-bootstrap interval of [-14.06,+1.56] points. A nonsignificant difference is not evidence of noninferiority; no hiring-utility preservation claim is made. A post-review analysis of the practical character-deletion baseline finds 168/192 correct as well. Its decisions differ from the compiled method in twelve cases, despite the same aggregate accuracy, and from native scoring in twenty-five cases. Thus score/encoding differences alone do not establish a practical utility advantage for either approach.

On the 100 official ARC-Challenge questions, both paths score 68/100. Every full-vocabulary array is identical. This is direct preservation on the tested all-evidence inputs. It does not cancel the credential-task loss or establish universal preserved performance.

The fidelity audit finds varying native scores in all thirty-two contexts. At least five binary native decisions must change under any name-invariant rule on these panels; the implemented route changes fifteen. Thus some changes are mathematically necessary for native fidelity, but this method is not the minimum-change invariant rule. Its average best-possible uniform logit fidelity error floor is 0.44242. Accuracy against the qualification labels is separate.

5.4 Balanced timing follow-up

The balanced follow-up contains eighty paired repetitions across eight families and two names. Internal/external median ratios are first aggregated within family; the median family ratio is 1.01118. The mean family ratio is 1.01179, with a descriptive family-bootstrap interval [1.01104,1.01266]. This is about 1.1% online overhead on this host, with concurrent load and setup/review costs excluded. It is not evidence that internal routing is faster than equivalent external routing. The initial dependent-case timing summary remains available but is not used for this efficiency assessment.

5.5 Reproducibility and review status

All public synthetic inputs are generated from published source. Per-case synthetic records, stage fingerprints, plans, analyses and tests are released. The benchmark source URL, subset IDs and snapshot hash are recorded. Earlier private real-CV experiments remain separate supporting studies and are not represented as independent validation of this new cohort.

Rebuilding the derivative in a new directory reproduces every pinned checkpoint file and the parameter digest, with all source files unchanged. A fresh-process repeat preserves all 192 synthetic rows and scoring-stage fingerprints and all 100 ARC records exactly. An initial reconstruction check incorrectly hashed a model whose arrays had been donated by serialization; offline tokenizer metadata also differed. The failed check and corrected save/reload procedure are retained. This is same-host reconstruction, not independent-laboratory replication.

A post-review service check repeats all 192 normal API comparisons against the actual timed external service and validates an explicit ordered 31-stage trace, expected shapes, finite arrays and the original fingerprints. This closes a bridge missing from the initial direct-kernel comparison.

Two fresh-context AI agents reviewed a blinded 24-case annotation packet before receiving methods or model results. They agreed on all identity spans, retained spans, qualification statuses and uncertainty decisions; both marked six cases uncertain. The packet repeats four sentence patterns across six names. Agreement is automated synthetic consistency evidence, not independent human accuracy. Their later technical reviews exposed comparator confounding, timing dependence, a span-fixture offset error, an incomplete cohort audit and semantic-support wording. The original reviews and response ledger are preserved. Human annotation reliability, real job-domain labels, a second model family and external-laboratory replication remain outstanding validation requirements.

6. Discussion and implications

The most useful contribution is a falsifiable contract: score equality is conditioned on exact evidence identity, and rejection is included in the audit. The compiler supplies a finite feasibility certificate instead of assuming that whitespace repair makes every native-token input comparable. The strong external baseline prevents attributing an identical-evidence property to internal location alone.

For the challenge brief, this addresses behavioral and scoring-activation bias from designated names: accepted variants have equal score-path states and outputs. Behavioral equality alone would not imply erasure; a hidden state can retain a name in a component the output ignores. Our named raw embeddings still distinguish names, deliberately preserving general model knowledge.

The sociotechnical risks include incorrect role annotation, unequal rejection, misleading confidence in a synthetic rubric, demographic proxies in retained qualifications and unfair job requirements. A model can be name-invariant while making consistently poor or unfair decisions. Human review, appeals and auditing broader group outcomes remain necessary for a hiring deployment; none is supplied by the score theorem.

7. Limitations and scope of submission

The study uses one checkpoint, one runtime, constructed task labels and a finite name panel. Its exact theorem depends on correct role review, native-compatible boundaries, fixed evidence, a 2,048-token input limit and deterministic trusted execution. It covers scalar scoring rather than unrestricted generated text. Side channels and raw embeddings remain visible to the trusted host. The compiler's optimality is finite and policy-relative; future-name acceptance is not guaranteed. Timing under concurrent load is descriptive.

The proposed mathematical contribution uses elementary established tools. Its scientific value must be assessed as a focused tokenizer-boundary and audit study. The work is suitable for external technical review as a transparent preprint draft; acceptance and venue fit are not established. A stronger hiring claim requires independent human annotations, credible real task labels and independent replication. Those cannot be supplied by additional tests on the same constructed data.

8. Availability, ethics and assistance

Code, public synthetic data, audit plans, aggregate and synthetic per-case records are included in nameguard-peer-review/. No actual applicant records, learned CV-derived adapter weights, reviewer secrets or base model weights are published. ARC source material is fetched separately under its dataset terms. Earlier empirical adapter failures remain available in the repository.

Substantial AI assistance was used in mathematical checking, code generation, analysis and manuscript drafting. The named author must verify the arguments, reproduce the experiments, confirm contributions and authorship, and complete venue-specific disclosures before submission. The blinded AI reviewers are identified as automated reviewers, not independent human participants.

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